A Formal Theory of Credibility: Why Assertions of Trustworthiness Decrease Trust

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Abstract

Credibility is a physical quantity. This paper grounds signaling theory in the thermodynamics of information processing, deriving computable bounds on rational belief update from Landauer's principle and Paper 4's Counting Gap.

Physical grounding. Signal production cost is measured in joules. By Landauer's principle, each irreversible bit operation costs at minimum $k_B T \ln 2$. A *cheap talk* signal has cost differential $\Delta = \text{Cost}(s \mid \bot) - \text{Cost}(s \mid \top) = 0$: the energy expenditure to produce the signal is truth-independent. A *costly signal* has $\Delta > 0$ in joules. Machine-checked proofs achieve $\Delta = \infty$: by Paper 4's Counting Gap ($\varepsilon \cdot N \leq C \Rightarrow N \leq C/\varepsilon$), the false-positive rate $\varepsilon_F \to 0$ of a sound type-checker implies that producing a false compiling proof requires traversing a diverging search space, hence diverging energy.

Theorem (Cheap Talk Bound). For any signal s with $\Delta = 0$, posterior credibility is bounded: $\Pr[C{=}1 \mid s] \leq p/(p + (1-p)q)$, where p is the prior and q is the mimicability of s. Text is cheap talk. All meta-assertions about credibility are cheap talk.

Theorem (Emphasis Penalty). Under the signaling equilibrium condition that deceptive agents produce repetitions at least as readily as honest agents, n repeated assertions give $\partial C(c, s_{1..n})/\partial n \leq 0$ for $n > k^*$. Additional assertions at $\Delta = 0$ carry no energy cost differential and therefore cannot increase rational credibility past the threshold.

Theorem (Text Credibility Bound). For high-magnitude claims (prior $p < e^{-M^*}$), no text string achieves credibility above $\tau < 1$. Rephrasing is energy-neutral; it cannot escape the bound.

Theorem (Costly Signal Escape). As $\Delta \to \infty$, $\Pr[C = 1 \mid s] \to 1$. Machine-checked proofs achieve this limit. The cost differential is grounded by the Counting Gap, not assumed.

Corollary (Counting Gap grounds $\Delta_{\textbf{Lean}}$). $\Delta_{\text{Lean}} = \infty$ follows from Paper 4, Theorem 3 (Counting Gap) and Theorem 13 (Energy Lower Bound). The impossibility of compiling a false proof is arithmetic, not a design choice.

Credibility leverage $L_C = \Delta C/\text{Signal Cost}$ is maximized by concentrating energy budget on high-Δ channels (proofs, demonstrations) and minimizing cheap talk. The theorems are formalized in Lean 4; all Lean proofs compile with 0 sorry.

Keywords: signaling theory, cheap talk, physical information theory, Landauer's principle, Counting Gap, Bayesian epistemology, costly signals, formal verification, Lean 4

1 Introduction

A puzzling phenomenon occurs in human and human-AI communication: emphatic assertions of trustworthiness often *reduce* perceived trustworthiness. "Trust me" invites suspicion. "I'm not lying" suggests deception. Excessive qualification of claims triggers doubt rather than alleviating it [13].

This paper provides the first formal framework for understanding this phenomenon. Our central thesis:

> **Credibility is bounded by signal cost. Assertions with truth-independent production costs cannot shift rational priors beyond computable thresholds.**

1.1 The Credibility Paradox

Observation: Let $C(s)$ denote credibility assigned to statement s. For assertions a about credibility itself:

$$\frac{\partial C(s \cup a)}{\partial |a|} < 0 \text{ past threshold } \tau$$

Adding more credibility-assertions *decreases* total credibility. This is counterintuitive under naive Bayesian reasoning but empirically robust, as explored in foundational models of reputation and trust [24, 13].

Examples: - "This is absolutely true, I swear" $<$ "This is true" $<$ stating the claim directly - Memory containing "verified, don't doubt, proven" triggers more skepticism than bare facts - Academic papers with excessive self-citation of rigor invite reviewer suspicion

1.2 Core Insight: Cheap Talk Bounds

The resolution comes from signaling theory [25, 7]. Define:

Cheap Talk: A signal s is *cheap talk* if its production cost is independent of its truth value: $\text{Cost}(s|\text{true}) = \text{Cost}(s|\text{false})$

Theorem (Informal): Cheap talk cannot shift rational priors beyond bounds determined by the prior probability of deception [12, 7].

Verbal assertions—including assertions about credibility—are cheap talk. A liar can say "I'm trustworthy" as easily as an honest person. Therefore, such assertions provide bounded evidence.

1.3 Connection to Leverage

This paper extends the leverage framework (Paper 3) [22] to epistemic domains. While Paper 4 characterizes the computational hardness of deciding which information to model [20], this paper characterizes the epistemic bounds of communicating that information.

Credibility Leverage: $L_C = \frac{\Delta\text{Credibility}}{\text{Signal Cost}}$

- Cheap talk: Cost ≈ 0, but ΔC bounded $\rightarrow L_C$ finite but capped
- Costly signals: Cost > 0 and truth-dependent $\rightarrow L_C$ can be unbounded
- Meta-assertions: Cost $= 0$, subject to recursive cheap talk bounds

1.4 Contributions

1. **Formal Framework (Section 2):** Rigorous definitions of signals, costs, credibility functions, and rationality constraints.

2. **Cheap Talk Theorems (Section 3):**

   - Theorem 3.1: Cheap Talk Bound
   - Theorem 3.2: Magnitude Penalty (credibility decreases with claim magnitude)
   - Theorem 3.3: Emphasis Penalty (excessive assertion decreases credibility)
   - Theorem 3.4: Meta-Assertion Trap (recursive bound on assertions about assertions)

3. **Costly Signal Characterization (Section 4):**

   - Definition of truth-dependent costs
   - Theorem 4.1: Costly signals can shift priors unboundedly
   - Theorem 4.2: Cost-credibility equivalence

4. **Impossibility Results (Section 5):**

   - Theorem 5.1: No string achieves credibility above threshold for high-magnitude claims
   - Corollary: Memory phrasing cannot solve credibility problems

5. **Leverage Integration (Section 6):** Credibility as DOF minimization; optimal signaling strategies.

6. **Machine-Checked Proofs (Appendix):** All theorems formalized in Lean 4 [9, 26].

Claim typing. As in Paper 4, strong claims are typed by regime rather than asserted globally: cheap-talk channel claims (`[CT]`), verified-signal claims (`[VS]`), and domain-specific audience claims (`[M]` mathematical, `[S]` social). Applied recommendations are valid only in the tagged regime where the theorem is proved.

1.5 Anticipated Objections

Before proceeding, we address objections readers are likely forming. Each is refuted in detail in Appendix 10.

"Signaling theory is old—this isn't novel." Spence's signaling model [25] and Crawford-Sobel's cheap talk [7] are foundational. Our contribution is (1) applying these to *meta-assertions* (claims about credibility), (2) proving *computable bounds* on credibility gain, and (3) integrating with the leverage framework. The theorems are new; the foundations are established.

"Real communication doesn't follow Bayesian rationality." The rationality assumption is an idealization that provides upper bounds. If agents deviate from Bayesian reasoning, credibility bounds may be tighter, not looser. The theorems characterize what is *achievable* under optimal reasoning—a ceiling, not a prediction.

"Costly signals aren't always truth-dependent." Correct. The definition (Section 2) distinguishes truth-dependent costs (credibility-enhancing) from truth-independent costs (not credibility-enhancing). Expensive signals that are equally costly for liars and truth-tellers remain cheap talk despite their cost.

"The magnitude penalty seems wrong—detailed claims are more credible." Detail *about the claim* can be credibility-enhancing (costly signal: research effort). Detail *about credibility itself* is cheap talk and subject to the magnitude penalty. The theorem distinguishes signal content from signal cost.

"The Lean proofs are just type-checking, not real mathematics." The Lean proofs formalize the mathematical structure of signaling theory. They verify that the cheap talk bounds follow from the definitions. The contribution is machine-checked precision, not computational complexity.

If you have an objection not listed above, check Appendix 10 before concluding it has not been considered.

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2 Foundations

2.1 Two Credibility Domains

This paper distinguishes two fundamentally different credibility domains that obey different dynamics:

Definition 2.0a (Mathematical Credibility). *Mathematical credibility* C_M measures the probability that a claim is logically sound. The audience is a formal verifier (proof assistant, compiler, test suite). The signal space consists of artifacts that can be mechanically checked. Mathematical credibility is binary at the limit: a proof compiles or it doesn't.

Definition 2.0b (Social Credibility). *Social credibility* C_S measures the probability that a social audience accepts a claim. The audience is human agents with priors shaped by institutional hierarchy, reputation, and group membership. The signal space consists of credentials, affiliations, endorsements, and communication patterns.

Theorem 2.0c (Domain Independence). Mathematical credibility and social credibility are orthogonal:

$$C_M(c,s) \not\Rightarrow C_S(c,s) \quad \text{and} \quad C_S(c,s) \not\Rightarrow C_M(c,s)$$

A signal maximizing one domain does not necessarily affect the other.

Proof. Constructive. (1) High C_M, low C_S: a correct proof by an unknown author receives $C_M \to 1$ (it compiles) but may receive $C_S \approx P_{\text{prior}}$ (no institutional endorsement). (2) High C_S, low C_M: a claim by a prestigious institution without formal verification receives high C_S (reputation transfer) but C_M is undefined or low (no proof). ■

Corollary 2.0d (Costly Signal Domain-Specificity). Costly signals are domain-specific:

- Machine-checked proofs are maximally costly in the mathematical domain (cannot compile a false proof) but may be cheap talk in the social domain (anyone can claim to have proofs).
- Institutional credentials (PhD, tenure, affiliation) are costly in the social domain (years of compliance) but cheap talk in the mathematical domain (credential $\not\Rightarrow$ soundness).

Remark (Credibility Domain Conflict). When a signal achieves $C_M \to 1$ but $C_S \approx 0$, the mathematical and social domains are in conflict. A rational response in the mathematical domain (engage with proofs) differs from a rational response in the social domain (defer to hierarchy). Observers may respond in either domain. This paper's theorems apply within each domain separately; cross-domain dynamics require modeling both simultaneously.

2.2 Signals and Costs

Definition 2.1 (Signal). A *signal* is a tuple $s = (c, v, p)$ where: c is the *content* (what is communicated), $v \in \{\top, \bot\}$ is the *truth value* (whether content is true), and $p \in \mathbb{R}_{\geq 0}$ is the *production cost*.

Physical grounding of cost. Production cost is measured in joules. By Landauer's principle [?], each irreversible bit operation erases at minimum

$$E_{\min} = k_B T \ln 2 \;\approx\; 2.8 \times 10^{-21} \text{ J at } T = 300\,\text{K},$$

where k_B is Boltzmann's constant. Paper 4's Energy Lower Bound (Theorem 13) shows that srank bits of structural information carry a minimum physical energy cost proportional to $\text{srank} \cdot k_B T \ln 2$. Signal production cost inherits this floor.

Definition 2.2 (Cheap Talk). A signal s is *cheap talk* if production cost is truth-independent:

$$\text{Cost}(s|v = \top) = \text{Cost}(s|v = \bot)$$

In physical units: the joules expended to produce s do not depend on whether $v = \top$ or $v = \bot$. Typed text is cheap talk: the Landauer cost of typing "true" equals the cost of typing "false."

Definition 2.3 (Costly Signal). A signal s is *costly* if:

$$\Delta := \text{Cost}(s|v = \bot) - \text{Cost}(s|v = \top) > 0$$

Producing the signal when false costs strictly more than when true. The quantity Δ (in joules) is the *cost differential*.

Definition 2.3a (Credibility as Perceived Integrity via Heuristic). A *credibility heuristic* is a polynomial-time computable function $h : \mathcal{C} \times \mathcal{S}^* \to [0, 1]$ that maps a claim and observed signals to a perceived integrity score. The *credibility* of a claim c given signals s is defined as:

$$C(c, s) := h(c, s)$$

where h is the receiver's heuristic for assessing claim integrity.

Rationale. This definition makes credibility operational: it is not an abstract notion but a *perceived* assessment computed by a specific *heuristic*. The receiver's cognitive system uses shortcuts to evaluate whether a claim's proponent is trustworthy, competent, or sincere. This grounds the subsequent analysis in algorithmic information processing rather than undefined intuition.

Heuristic Properties. We assume h satisfies: (i) $h(c, \emptyset)$ returns the receiver's prior (baseline integrity belief), (ii) h is efficient (bounded computation), and (iii) h may exhibit systematic bias (deviation from Bayesian optimality).

Mathematical vs Social Domain. The heuristic differs by domain:

- h_M: mathematical credibility heuristic (type-checker, proof verifier) - converges to 0/1 as verification runs (bounded verification $\to$ bounded error)

- h_S: social credibility heuristic (reputation, credentials, rhetorical style) - bounded away from extremes due to cognitive limitations (bounded memory $\to$ bounded precision)

Remark (Lean proofs as physical signals). A machine-checked Lean proof is a costly signal with $\Delta \to \infty$. To produce a compiling proof of a false theorem, the type-checker's false-positive rate satisfies $\varepsilon_F \to 0$ (soundness). By Paper 4's Counting Gap (Theorem 3): $\varepsilon \cdot N \leq C \Rightarrow N \leq C/\varepsilon$. With $\varepsilon_F \to 0$, the number of false proofs that pass verification is bounded by $C \cdot \varepsilon_F \to 0$. Producing one would require traversing a search space of diverging size, hence diverging energy. Therefore $\Delta_{\text{Lean}} = \infty$: the Lean type-checker physically separates true from false.

Intuition: Verbal assertions are cheap talk—saying "I'm honest" costs the same whether you're honest or not. A PhD from MIT is a costly signal [25]—obtaining it while incompetent is much harder than while competent. Similarly, price and advertising can serve as signals of quality [17].

2.3 Credibility Functions

Definition 2.4 (Prior). A *prior* is a probability distribution $P : \mathcal{C} \to [0,1]$ over claims, representing beliefs before observing signals.

Definition 2.5 (Credibility Function). A *credibility function* is a mapping:

$$C : \mathcal{C} \times \mathcal{S}^* \to [0,1]$$

from (claim, signal-sequence) pairs to credibility scores, satisfying: 1. $C(c, \emptyset) = P(c)$ (base case: prior) 2. Bayesian update: $C(c, s_{1..n}) = P(c|s_{1..n})$

Definition 2.6 (Rational Agent). An agent is *rational* if: 1. Updates beliefs via Bayes' rule 2. Has common knowledge of rationality [3] (knows others are rational, knows others know, etc.) 3. Accounts for strategic signal production [6].

2.4 Deception Model

Definition 2.7 (Deception Prior). Let $\pi_d \in [0,1]$ be the prior probability that a random agent will produce deceptive signals. This is common knowledge.

Definition 2.8 (Magnitude). The *magnitude* of a claim c is:

$$M(c) = -\log P(c)$$

High-magnitude claims have low prior probability. This is the standard self-information measure [19].

2.5 Model Contract and Regime Tags

All theorem statements in this paper are typed by the following contract:

- **K1 (binary claim state):** claim truth is modeled as $C \in \{0,1\}$.
- **K2 (Bayesian receiver):** posterior credibility is computed by Bayes updates.
- **K3 (signal channel declaration):** each theorem declares whether the signal channel is cheap talk or verifier-backed.
- **K4 (audience domain declaration):** each theorem declares mathematical-domain (C_M) or social-domain (C_S) scope.

We use the following regime tags:

- `[CT]`: cheap-talk channel (truth-independent signal cost, $\Delta = 0$),

- [VS]: verifier-backed signal channel ($\varepsilon_T, \varepsilon_F$ model, $\Delta \to \infty$),
- [M]: mathematical credibility domain,
- [S]: social credibility domain.

Proposition 2.1 (Declared Regime Coverage for Credibility Claims). *Each declared regime above has at least one theorem-level mechanized core in Lean: [CT] via* `cheap_talk_bound`, `magnitude_penalty`, `emphasis_penalty`*; [VS] via* `verified_signal_credibility`, `proof_as_ultimate_signal`*; [M]/[S] via* `domain_independence_math_not_implies_social` *and* `domain_independence_social_not_implies_math`*. The physical connection between [CT] ($\Delta = 0$) and [VS] ($\Delta \to \infty$) is grounded in Landauer's principle and Paper 4's Counting Gap (Theorem 3).*

Proof. Direct by inspection of the theorem declarations in `Credibility/CheapTalk.lean`, `Credibility/CostlySignals.lean`, and `Credibility/Basic.lean`. ■

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3 Cheap Talk Theorems

3.1 The Cheap Talk Bound

Theorem 3.1 (Cheap-talk credibility is a likelihood-ratio bound). *Let $C \in \{0,1\}$ denote the truth of a claim ($C = 1$ true), with prior $p := \Pr[C = 1] \in (0,1)$. Let S be the event that the receiver observes a particular message-pattern (signal) s.*

C21 C1

Define the emission rates

$$\alpha := \Pr[S \mid C = 1], \qquad \beta := \Pr[S \mid C = 0].$$

Then the posterior credibility of the claim given observation of s is

$$\Pr[C = 1 \mid S] \;=\; \frac{p\,\alpha}{p\,\alpha + (1-p)\,\beta}.$$

Equivalently, in odds form,

$$\frac{\Pr[C = 1 \mid S]}{\Pr[C = 0 \mid S]} \;=\; \frac{p}{1-p} \cdot \frac{\alpha}{\beta}.$$

In particular, if s is a cheap-talk *pattern in the sense that:*

(i) truthful senders emit s with certainty ($\alpha = 1$), and

(ii) deceptive senders can mimic s with probability at least q (i.e. $\beta \geq q$),

then credibility obeys the tight upper bound

$$\Pr[C = 1 \mid S] \;\leq\; \frac{p}{p + (1-p)q}.$$

Moreover this bound is tight*: equality holds whenever $\alpha = 1$ and $\beta = q$.*

Proof. By Bayes' rule,

$$\Pr[C = 1 \mid S] = \frac{\Pr[S \mid C = 1] \Pr[C = 1]}{\Pr[S \mid C = 1] \Pr[C = 1] + \Pr[S \mid C = 0] \Pr[C = 0]} = \frac{p\alpha}{p\alpha + (1-p)\beta}.$$

If $\alpha = 1$ and $\beta \geq q$, the denominator is minimized by setting $\beta = q$, yielding

$$\Pr[C = 1 \mid S] \leq \frac{p}{p + (1-p)q}.$$

Tightness is immediate when $\beta = q$. ■

Remark (Notation reconciliation). In this paper we use q to denote the *mimicability* of a cheap-talk signal: the probability that a deceptive sender successfully produces the same message pattern as a truthful sender. If one prefers to work with detection probability π_d (the probability deception is detected), then $q = 1 - \pi_d$ and the bound becomes $\Pr[C = 1 \mid S] \leq p/(p + (1-p)(1-\pi_d))$.

Interpretation: No matter how emphatically you assert something, cheap talk credibility is capped. The cap depends on how likely deception is in the population.

3.2 The Magnitude Penalty

Theorem 3.2 (Magnitude Penalty). *For claims c_1, c_2 with $M(c_1) < M(c_2)$ (i.e., $p_1 := P(c_1) > p_2 := P(c_2)$) and identical cheap talk signals s with mimicability q:*

$$\Pr[c_1 \mid S] > \Pr[c_2 \mid S]$$

Higher-magnitude claims receive less credibility from identical signals.

C5 C2

Proof. From Theorem 3.1, the bound $p/(p + (1-p)q)$ is strictly increasing in p for fixed $q \in (0, 1)$. Since $p_1 > p_2$, we have $\Pr[c_1 \mid S] > \Pr[c_2 \mid S]$. ■

Interpretation: Claiming you wrote one good paper gets more credibility than claiming you wrote four. The signal (your assertion) is identical; the prior probability differs.

3.3 The Emphasis Penalty

Theorem 3.3 (Emphasis Penalty). *Let $s_1, s_2, \ldots, s_n$ be cheap talk signals all asserting claim c with the same mimicability q and prior $p = \Pr[C = 1]$. Assume the signaling equilibrium condition:*

(EP) Deceptive agents produce repetitions at least as readily as honest agents: $\Pr[s_k \mid C{=}0,\, s_{1..k-1}] \geq \Pr[s_k \mid C{=}1,\, s_{1..k-1}]$ for $k > k^*$.

Then for $n > k^*$: $\frac{\partial C(c, s_{1..n})}{\partial n} \leq 0$, with strict decrease whenever the inequality in (EP) is strict.

C4 C9 C3

Proof. Let $p_k := \Pr[C = 1 \mid s_1, \ldots, s_k]$ denote the posterior after k assertions. By Bayes' rule applied to the k-th update:

$$p_k = \frac{p_{k-1}\, \alpha_k}{p_{k-1}\, \alpha_k + (1 - p_{k-1})\, \beta_k}$$

where $\alpha_k := \Pr[s_k \mid C{=}1, s_{1..k-1}]$ and $\beta_k := \Pr[s_k \mid C{=}0, s_{1..k-1}]$.

The odds form is:

$$\frac{p_k}{1 - p_k} = \frac{p_{k-1}}{1 - p_{k-1}} \cdot \frac{\alpha_k}{\beta_k}.$$

Under condition (EP), $\alpha_k/\beta_k \leq 1$ for $k > k^*$. Therefore the odds ratio is non-increasing, i.e., $p_k \leq p_{k-1}$ for $k > k^*$. When (EP) holds with strict inequality, $p_k < p_{k-1}$, giving $\partial C/\partial n < 0$.

Physical interpretation. Since all assertions are cheap talk ($\Delta = 0$ in joules), additional assertions carry zero energy cost differential. A rational receiver therefore assigns the repetition count n the same analysis as any cheap-talk signal: it is informative only about the sender's strategy. Under common knowledge of rationality, deceptive senders choose large n to mimic honest senders; receivers anticipate this, making (EP) the equilibrium condition. ■

Interpretation: "Trust me, I'm serious, this is absolutely true, I swear" is *less* credible than just stating the claim. The emphasis signals desperation.

3.4 The Meta-Assertion Trap

Theorem 3.4 (Meta-Assertion Trap). *Let a be a cheap talk assertion and m be a meta-assertion "assertion a is credible." Then:*

$$C(c, a \cup m) \leq C(c, a) + \epsilon$$

where $\epsilon \to 0$ as common knowledge of rationality increases.

C8 C7 C6

Proof. Meta-assertion m is itself cheap talk (costs nothing to produce regardless of truth). Therefore m is subject to the Cheap Talk Bound (Theorem 3.1).

Under common knowledge of rationality, agents anticipate that deceptive agents will produce meta-assertions. Therefore:

$$P(m|\text{deceptive}) \approx P(m|\text{honest})$$

The signal provides negligible information; $\epsilon \to 0$. ■

Interpretation: "My claims are verified" is cheap talk about cheap talk. It doesn't escape the bound—it's *subject to* the bound recursively. Adding "really verified, I promise" makes it worse.

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4 Costly Signal Characterization

4.1 Definition and Properties

Theorem 4.1 (Costly Signal Effectiveness). *For costly signal s with cost differential $\Delta = Cost(s|\bot) - Cost(s|\top) > 0$:*

$$\Pr[C = 1 \mid S] \to 1 \textit{ as } \Delta \to \infty$$

Costly signals can achieve arbitrarily high credibility.

C10 C14

Proof. If Δ is large, deceptive agents cannot afford to produce s, so $\beta := \Pr[S \mid C = 0] \to 0$ as $\Delta \to \infty$. Applying Theorem 3.1 with $\alpha = 1$:

$$\Pr[C = 1 \mid S] = \frac{p}{p + (1-p)\beta} \to 1 \text{ as } \beta \to 0.$$

■

Theorem 4.2 (Verified signals drive credibility to 1). *Let* $C \in \{0,1\}$ *with prior* $p = \Pr[C = 1]$. *Suppose a verifier produces an acceptance event* A *such that*

$$\Pr[A \mid C = 1] \geq 1 - \varepsilon_T, \qquad \Pr[A \mid C = 0] \leq \varepsilon_F,$$

for some $\varepsilon_T, \varepsilon_F \in [0,1]$. *Then*

$$\Pr[C = 1 \mid A] \geq \frac{p(1-\varepsilon_T)}{p(1-\varepsilon_T) + (1-p)\varepsilon_F}.$$

In particular, if $\varepsilon_F \to 0$ *and* ε_T *is bounded away from* 1*, then* $\Pr[C = 1 \mid A] \to 1$.
C13 C12

Proof. Apply Theorem 3.1 with $S := A$, $\alpha := \Pr[A \mid C = 1]$, $\beta := \Pr[A \mid C = 0]$, then use $\alpha \geq 1-\varepsilon_T$ and $\beta \leq \varepsilon_F$. ■

Remark. This theorem provides the formal bridge to machine-checked proofs: Lean corresponds to a verifier where false claims have negligible acceptance probability ($\varepsilon_F \approx 0$, modulo trusted kernel assumptions). The completeness gap ε_T captures the effort to construct a proof.

4.2 Examples of Costly Signals

Signal	Cost if True	Cost if False	Credibility Shift
PhD from MIT	4 years effort	4 years + deception risk	Moderate
Working code	Development time	Same + it won't work	High
Verified Lean proofs	Proof effort	Impossible (won't compile)	Maximum
Verbal assertion	~0	~0	Bounded

Key insight: Lean proofs with `0 sorry` are *maximally costly signals.* You cannot produce a compiling proof of a false theorem. The cost differential is infinite [9, 8].

Theorem 4.3 (Proof as Ultimate Signal). *Let* s *be a machine-checked proof of claim* c*. Then:*

$$\Pr[c \mid s] = 1 - \varepsilon$$

where ε *accounts only for proof assistant bugs.*
C11 C14

Proof. This is a special case of Theorem 4.2 with $\varepsilon_T \approx 0$ (proof exists if claim is true and provable) and $\varepsilon_F \approx 0$ (proof assistant soundness). See [9, 8]. ■

Corollary 4.4 (Counting Gap grounds $\Delta_{\text{Lean}} = \infty$). *The infinite cost differential of a machine-checked proof is a consequence of Paper 4's Counting Gap (Theorem 3:* $\varepsilon \cdot N \leq C \Rightarrow N \leq C/\varepsilon$*) applied to the type-checker channel.*

Proof. Let N = number of false proofs that compile, C = capacity of the type-checker channel, $\varepsilon = \varepsilon_F$ = false-positive rate. The Counting Gap gives $N \le C/\varepsilon_F$. By soundness of Lean's kernel [9], $\varepsilon_F \to 0$, so $N \to 0$. To produce a single false proof that compiles, an agent must either: (a) find a kernel bug (probability $\varepsilon_F \to 0$), or (b) search the space of all type-correct proof terms for one that proves a false theorem. Space (b) has size at most $C/\varepsilon_F \to \infty$, requiring diverging computational work. By Paper 4's Energy Lower Bound (Theorem 13), computational work translates to physical energy at rate $\ge k_B T \ln 2$ per bit. Therefore $\Delta_{\text{Lean}} = \text{Cost}(s \mid \bot) - \text{Cost}(s \mid \top) = \infty$ in joules. The cost differential is not a modeling assumption; it follows from counting. ■

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5 Impossibility Results

5.1 The Text Credibility Bound

Theorem 5.1 (Text Credibility Bound). *For any text string T (memory content, assertion, etc.) and high-magnitude claim c with $M(c) > M^*$ (i.e., prior $p < e^{-M^*}$):*

$$\Pr[c \mid T] < \tau$$

where $\tau < 1$ is determined by the mimicability q and M^*. No text achieves full credibility for exceptional claims.

`C17 C15 C16`

Proof. Text is cheap talk (production cost independent of truth). Apply Theorem 3.1 with prior $p = e^{-M^*}$ and mimicability q:

$$\tau = \frac{p}{p + (1-p)q} = \frac{e^{-M^*}}{e^{-M^*} + (1 - e^{-M^*})q}$$

For M^* large (low prior probability), $\tau \to 0$ regardless of $q > 0$. ■

Corollary 5.2 (Memory Iteration Futility). No rephrasing of memory content can achieve credibility above τ for high-magnitude claims. Iteration on text is bounded in effectiveness.

Interpretation: This is why we couldn't solve the credibility problem by editing memory text. The *structure* of the problem (text is cheap talk, claims are high-magnitude) guarantees bounded credibility regardless of phrasing.

5.2 Optimal Strategies

Theorem 5.2 (Optimal Credibility Strategy). *For high-magnitude claims in regime `[CT]+[VS]`, an optimal strategy is:*

1. *Minimize cheap talk (reduce emphasis and meta-assertions),*
2. *Maximize costly-signal exposure (show work and verifiable artifacts),*
3. *Enable costly-to-fake demonstration channels.*

(Lean anchor: `optimal_strategy_dominance`.)

Proof. From Theorem 5.1, text-only updates are bounded for high-magnitude claims. From Theorem 4.2, verifier-backed channels strictly improve achievable posterior credibility as false-positive rate decreases. Therefore optimal policy minimizes bounded channels and allocates effort to verifier-backed costly channels. ∎

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6 Leverage Integration

6.1 Credibility as DOF Minimization

Applying the leverage framework (Paper 3) [22]:

Signal DOF: Words in an assertion are degrees of freedom. Each word can be independently modified.

Signal Leverage: $L_S = \frac{\Delta C}{\text{Words}}$

Theorem 6.1 (Credibility Leverage). *For cheap-talk signals with nonnegative credibility impact, leverage is maximized by minimizing word count:*

$$\arg\max_s L_S(s) = \arg\min_s |s|$$

subject to conveying the claim.

`C18` `C19` `C20`

Proof. With impact fixed and nonnegative, $L_S = \Delta C/\text{Words}$ is inverse-monotone in word count. Hence shorter valid signals weakly dominate longer ones in leverage. ∎

Interpretation: Shorter, terser memory entries achieve higher credibility leverage than verbose explanations. "70k lines, deployed in 3 labs" beats lengthy justification.

6.2 Energy Budget Interpretation

The cost differential Δ (in joules) induces a natural budget interpretation. Let B be an agent's total energy budget for signal production. Then:

- A cheap-talk strategy exhausts B producing signals with $\Delta = 0$, yielding credibility bounded by Theorem 3.1 regardless of B.
- A costly-signal strategy concentrates B on signals with large Δ, yielding credibility approaching 1 as $\Delta \to \infty$ (Theorem 4.1).
- Optimal credibility leverage (Theorem 6.1) therefore allocates budget entirely to high-Δ channels: *demonstrations, working code, machine-checked proofs* rather than assertions.

Corollary (Minimality of honest signaling). An honest agent with true claim c achieves maximal L_C by producing the minimal proof of c: one correctly compiling Lean file, zero additional assertions. Additional words add DOF (increase $|s|$) without increasing Δ, strictly decreasing leverage.

---

7 Related Work

Signaling Theory: Spence (1973) [25] introduced costly signaling in job markets. Zahavi (1975) [27] applied it to biology (handicap principle). Akerlof (1970) [1] established the foundational role of asymmetric information in market collapse. We formalize and extend to text-based communication.

Cheap Talk: Crawford & Sobel (1982) [7] analyzed cheap talk in game theory. Farrell (1987) [11] and Farrell & Rabin (1996) [12] further characterized the limits of unverified communication. We prove explicit bounds on credibility shift.

Epistemic Logic: Hintikka (1962) [14], Fagin et al. (1995) [10] formalized knowledge and belief. We add signaling structure.

Bayesian Persuasion: Kamenica & Gentzkow (2011) [15] studied optimal information disclosure. Our impossibility results complement their positive results.

Social Epistemology: Goldman (1999) [?] and Hardwig (1991) [?] studied the social dimensions of knowledge. Our dual truth framework extends this to include ego-driven truth as a complement to epistemic truth.

Cognitive Dissonance: Festinger (1957) [?] introduced cognitive dissonance theory, which explains how individuals resolve conflicts between beliefs and actions. Our coherence measure quantifies this dissonance as a gap between epistemic and ego-driven truth.

Dual Process Theories: Kahneman (2011) [?] and Evans (2003) [?] distinguished between fast, intuitive thinking (System 1) and slow, deliberate thinking (System 2). Our dual truth framework aligns with this distinction: ego-driven truth often operates through System 1, while epistemic truth requires System 2 reasoning.

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8 Conclusion

We have formalized why assertions of credibility can decrease perceived credibility, proved impossibility bounds on cheap talk, and characterized the structure of costly signals.

Key results: 1. Cheap talk credibility is bounded (Theorem 3.1) 2. Emphasis decreases credibility past threshold (Theorem 3.3) 3. Meta-assertions are trapped in the same bound (Theorem 3.4) 4. No text achieves full credibility for exceptional claims (Theorem 5.1) 5. Only costly signals (proofs, demonstrations) escape the bound (Theorem 4.1)

Implications: - Memory phrasing iteration has bounded effectiveness - Real-time demonstration is the optimal credibility strategy - Lean proofs are maximally costly signals (infinite cost differential)

Methodology and Disclosure

Role of LLMs in this work. This paper was developed through human-AI collaboration, and this disclosure is particularly apropos given the paper's subject matter. The author provided the core intuitions—the cheap talk bound, the emphasis paradox, the impossibility of achieving full credibility via text—while large language models (Claude, GPT-4) served as implementation partners for formalization, proof drafting, and LaTeX generation.

The Lean 4 proofs (2326 lines, 0 sorry placeholders) were iteratively developed: the author specified theorems, the LLM proposed proof strategies, and the Lean compiler verified correctness.

What the author contributed: The credibility framework itself, the cheap talk bound conjecture, the emphasis penalty insight, the connection to costly signaling theory, and the meta-observation that Lean proofs are maximally costly signals.

What LLMs contributed: LaTeX drafting, Lean tactic suggestions, Bayesian calculation assistance, and prose refinement.

Meta-observation: This paper was produced via the methodology it describes—intuition-driven, LLM-implemented—demonstrating in real-time the credibility dynamics it formalizes. The LLM-generated text is cheap talk; the Lean proofs are costly signals. The proofs compile; therefore the theorems are true, regardless of how the proof text was generated. This is the paper's own thesis applied to itself.

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9 Appendix: Lean Formalization

9.1 Verification Scope

All theorem-level claims in this paper are mapped to Lean declarations in `docs/papers/paper5_credibility/proofs/Credibility/*.lean`. The proof artifact is checked by running:

```
cd docs/papers/paper5_credibility/proofs
lake build
```

9.2 Module Structure

```
Credibility/
|- Basic.lean -- core definitions + domain-independence lemmas
|- CheapTalk.lean -- Theorems 3.1-3.4
|- CostlySignals.lean -- Theorems 4.1-4.4
|- Impossibility.lean -- Theorems 5.1-5.3 (+ asymptotic form)
|- Leverage.lean -- leverage theorems
`- CoherentStopping.lean -- stopping/threshold bridge utilities
```

9.3 Claim-to-Proof Mapping

Paper Claim	**Lean Handle(s)**	**Tag**
Theorem 3.1	`cheap_talk_bound`, `cheap_talk_bound_tight`	`[CT]`
Theorem 3.2	`magnitude_penalty`	`[CT]`
Theorem 3.3	`emphasis_penalty`	`[CT]`
Theorem 3.4	`meta_assertion_trap`, `meta_assertion_bounded`	`[CT]`
Theorem 4.1	`costly_dominates_cheap`	`[VS]`
Theorem 4.2	`verified_signal_credibility`, `verified_signal_limit_one`	`[VS]`
Theorem 4.3	`proof_as_ultimate_signal`	`[VS]`, `[M]`
Theorem 5.1	`text_credibility_bound`, `asymptotic_impossibility`	`[CT]`

Paper Claim	Lean Handle(s)	Tag
Theorem 5.2	`optimal_strategy_dominance`	[CT]+[VS]
Theorem 6.1	`credibility_leverage_minimization`, `brevity_principle`	[CT]
Proposition 2.1	`domain_independence_math_not_implies_social`, `domain_independence_social_not_implies_math`, `machine_proof_domain_specificity`	[M],[S]

9.4 Current Proof Statistics

Lean files: 13
Lean lines: 2326
Theorem/lemma statements: 82
Sorry placeholders: 0

---

10 Preemptive Rebuttals

We address anticipated objections to the credibility framework.

10.1 Objection 1: "Signaling theory is old—this isn't novel"

Objection: "Spence's signaling model and Crawford-Sobel's cheap talk are decades old. This paper just applies existing theory."

Response: The foundations are established; the application is novel. Our contributions:

1. **Meta-assertions:** We apply cheap talk bounds to *claims about credibility itself*—a recursive structure not analyzed in prior work
2. **Computable bounds:** We derive explicit formulas for credibility ceilings as functions of prior deception probability
3. **Leverage integration:** We connect credibility to the DOF framework, showing credibility as a form of epistemic leverage

The theorems (Cheap Talk Bound, Magnitude Penalty, Meta-Assertion Trap) are new. The methodology is established.

10.2 Objection 2: "Real communication isn't Bayesian"

Objection: "Humans don't update beliefs according to Bayes' rule. The rationality assumption is unrealistic."

Response: The rationality assumption provides *upper bounds*. If agents deviate from Bayesian reasoning, credibility bounds may be tighter (irrational skepticism) or looser (irrational credulity). The theorems characterize what is *achievable* under optimal reasoning—a ceiling, not a prediction.

For AI systems designed to be rational, the bounds are prescriptive. For human communication, they are normative benchmarks.

10.3 Objection 3: "Costly signals aren't always truth-dependent"

Objection: "Expensive signals can be equally costly for liars and truth-tellers. Your distinction is too clean."

Response: Correct. The definition (Section 2) distinguishes:

- **Truth-dependent costs:** Lying is more expensive than truth-telling (e.g., maintaining consistent false memories)
- **Truth-independent costs:** Equal cost regardless of truth value (e.g., verbose phrasing)

Expensive signals with truth-independent costs remain cheap talk despite their expense. The credibility-enhancing property comes from the *differential*, not the absolute cost.

10.4 Objection 4: "The magnitude penalty seems wrong"

Objection: "Detailed claims are often more credible, not less. The magnitude penalty contradicts intuition."

Response: The distinction is between:

- **Detail about the claim:** Research, evidence, specificity—these are costly signals (effort-dependent) and credibility-enhancing
- **Detail about credibility itself:** "I'm absolutely certain, trust me, this is verified"—these are cheap talk and subject to the magnitude penalty

The theorem applies to meta-assertions (claims about credibility), not to substantive claims. A detailed scientific argument is credibility-enhancing; a detailed assertion of trustworthiness is credibility-reducing.

10.5 Objection 5: "The Lean proofs are trivial"

Objection: "The Lean proofs just formalize definitions. There's no deep mathematics."

Response: The value is precision, not difficulty. The proofs verify:

1. The cheap talk bound follows from the cost-independence definition
2. The magnitude penalty follows from the recursive structure of meta-assertions
3. The impossibility results follow from the bound composition

Machine-checked proofs eliminate ambiguity in informal arguments. The contribution is verification, not complexity.

10.6 Objection 6: "This doesn't apply to AI systems"

Objection: "AI systems can be designed to be trustworthy. The cheap talk bounds don't apply to engineered systems."

Response: The bounds apply to *communication about* trustworthiness, not to trustworthiness itself. An AI system can be trustworthy, but its *assertions* of trustworthiness are cheap talk unless backed by costly signals (audits, formal verification, track record).

The framework explains why "I am safe and helpful" is less credibility-enhancing than demonstrated safety and helpfulness.

10.7 Objection 7: "The impossibility results are too strong"

Objection: "Theorem 5.1 says no string achieves high credibility for high-magnitude claims. But some claims are believed."

Response: The theorem applies to *cheap talk* strings. Credibility for high-magnitude claims requires costly signals:

- Track record (historical accuracy)
- Formal verification (mathematical proof)
- Reputation stake (costly to lose)
- Third-party attestation (independent verification)

The impossibility is for *verbal assertions alone*. Credibility is achievable through costly signals, not through phrasing.

10.8 Objection 8: "The leverage integration is forced"

Objection: "Connecting credibility to DOF seems like a stretch. These are different concepts."

Response: The connection is structural. Credibility leverage is:

$$L_C = \frac{\Delta\text{Credibility}}{\text{Signal Cost}}$$

This parallels architectural leverage:

$$L = \frac{|\text{Capabilities}|}{\text{DOF}}$$

Both measure efficiency: capability per unit of constraint. The integration shows that credibility optimization follows the same mathematical structure as architectural optimization. This unification is the theoretical contribution.

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